



## Checking and further securing understanding of the commutative law

### Task A: Reviewing product of prime factors

1) Write the following as a product of prime factors, without using a calculator.

a) 200  $2^3 \times 5^2$

d) 308  $2^2 \times 7 \times 11$

b) 196  $2^2 \times 7^2$

e) 23 400  $2^3 \times 3^2 \times 5^2 \times 13$

c) 378  $2 \times 3^3 \times 7$

2) Use  $60 = 2^2 \times 3 \times 5$  to write the following, without using a calculator, as a product of prime factors.

a) 120  $= 60 \times 2 = 2^2 \times 3 \times 5 \times 2 = 2^3 \times 3 \times 5$

b) 180  $= 60 \times 3 = 2^2 \times 3 \times 5 \times 3 = 2^2 \times 3^2 \times 5$

c) 300  $= 60 \times 5 = 2^2 \times 3 \times 5 \times 5 = 2^2 \times 3 \times 5^2$

d) 600  $= 60 \times 10 = 2^2 \times 3 \times 5 \times 2 \times 5 = 2^3 \times 3 \times 5^2$

e) 6000  $= 60 \times 100 = 2^2 \times 3 \times 5 \times 2^2 \times 5^2 = 2^4 \times 3 \times 5^3$

f) 4200  $= 60 \times 70 = 2^2 \times 3 \times 5 \times 7 \times 10 = 2^2 \times 3 \times 5 \times 7 \times 2 \times 5$

g) 30  $= 60 \div 2 = 2^2 \times 3 \times 5 \div 2 = 2 \times 3 \times 5$

**Task B: Using product of prime factors**

$$42 = 2 \times 3 \times 7$$

$$308 = 2^2 \times 7 \times 11$$

$$35 = 5 \times 7$$

$$195 = 3 \times 5 \times 13$$

$$90 = 2 \times 3^2 \times 5$$

$$198 = 2 \times 3^2 \times 11$$

1) Use the prime factors above to calculate the following:

a)  $42 \times 90$

$$3780$$

b)  $198 \times 42$

$$8316$$

c)  $198 \times 35$

$$6930$$

d)  $90 \times 308$

$$27\,720$$

e)  $195 \times 198$

$$38\,610$$